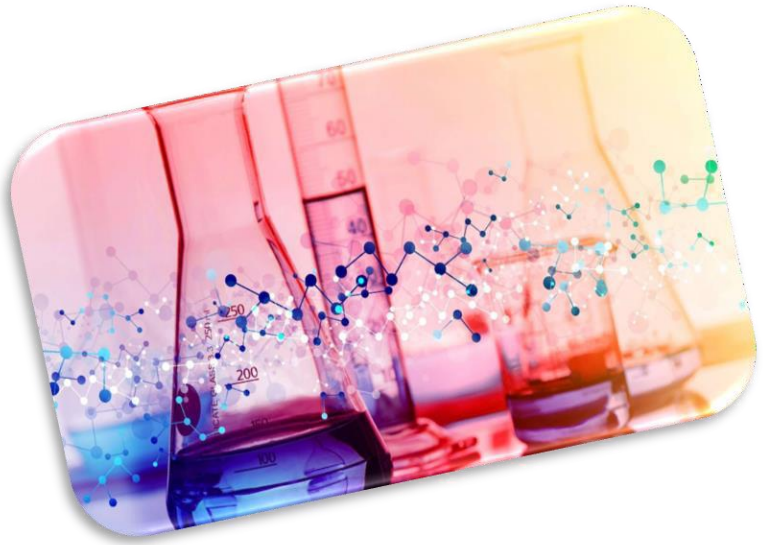
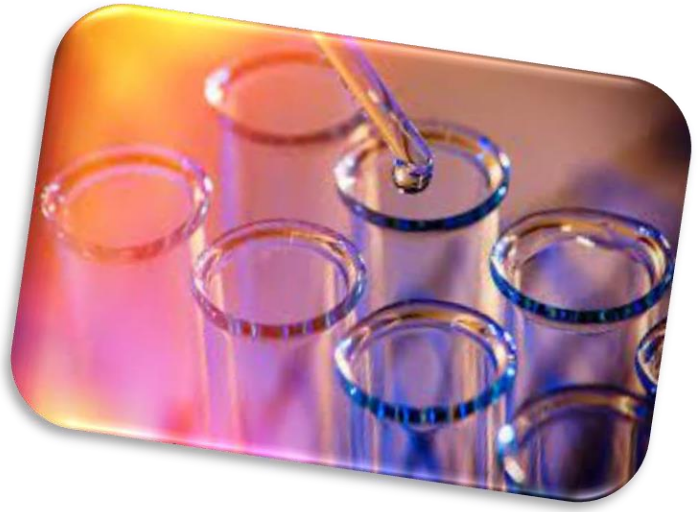
Arab Academy
for Science, Technology & Maritime Transport

Biochemistry
EBA1300





Dr. Mohamed R. Eletmany

PhD in Fiber Science and Polymer Chemistry

NC State Wilson College of Textiles, NC State University, North Carolina, USA

Lecture 6

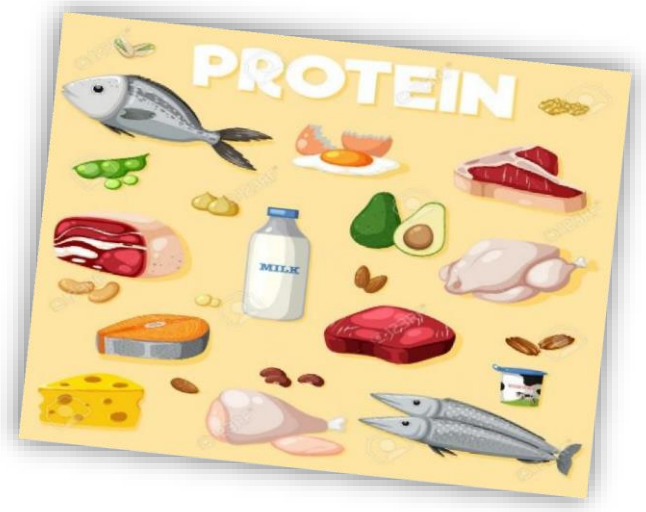
Proteins (part 1)

What are Proteins

- Proteins are the most common molecules in living systems.
- They are found in almost every cell in the body.
- Proteins are large molecules with large molecular weight.
- They have many different functions and required in every process in the body.



Proteins



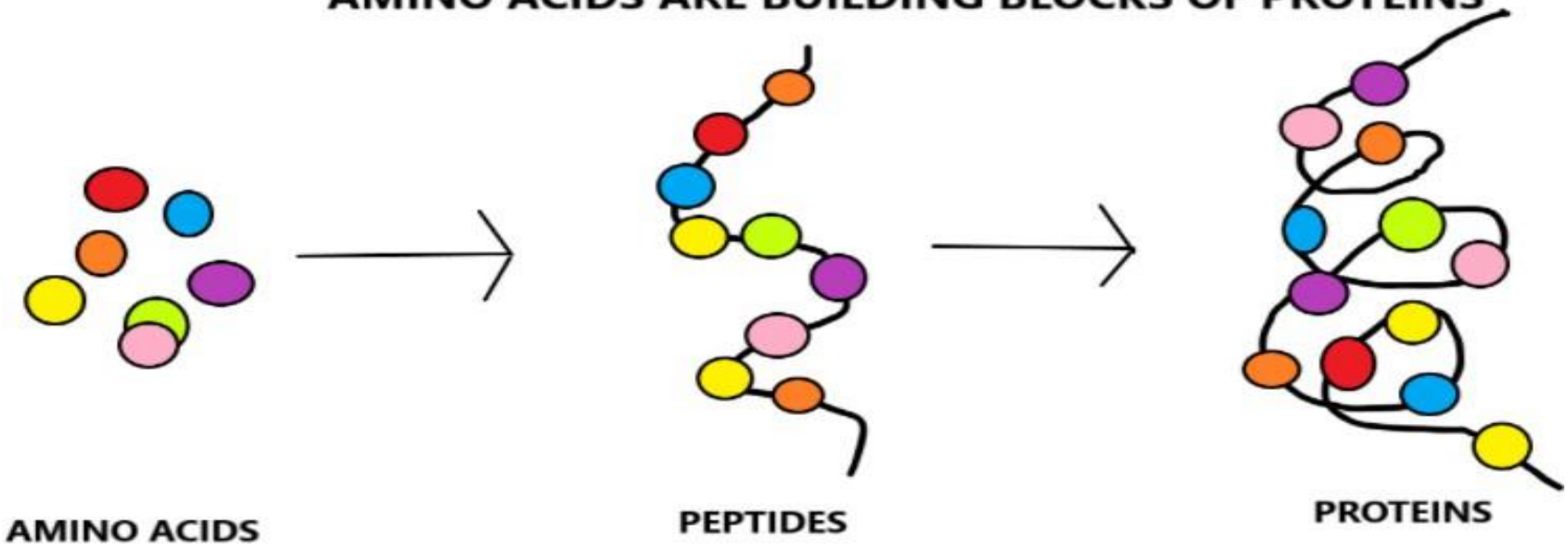
- They are important for growth and maintenance of the body.
- We can say that proteins are the building blocks of the life.
- Dietary sources of proteins are milk, cheese, fish, meat, etc.
- The building block of protein is called amino acid.

Proteins

- Proteins are polymers of amino acids.
- We can consider amino acids as monomers of proteins.
- Hydrolysis of proteins yields amino acids.
- Amino acid and protein structure is based on Carbon, Hydrogen, Oxygen and Nitrogen.

  (C, H, N, O)

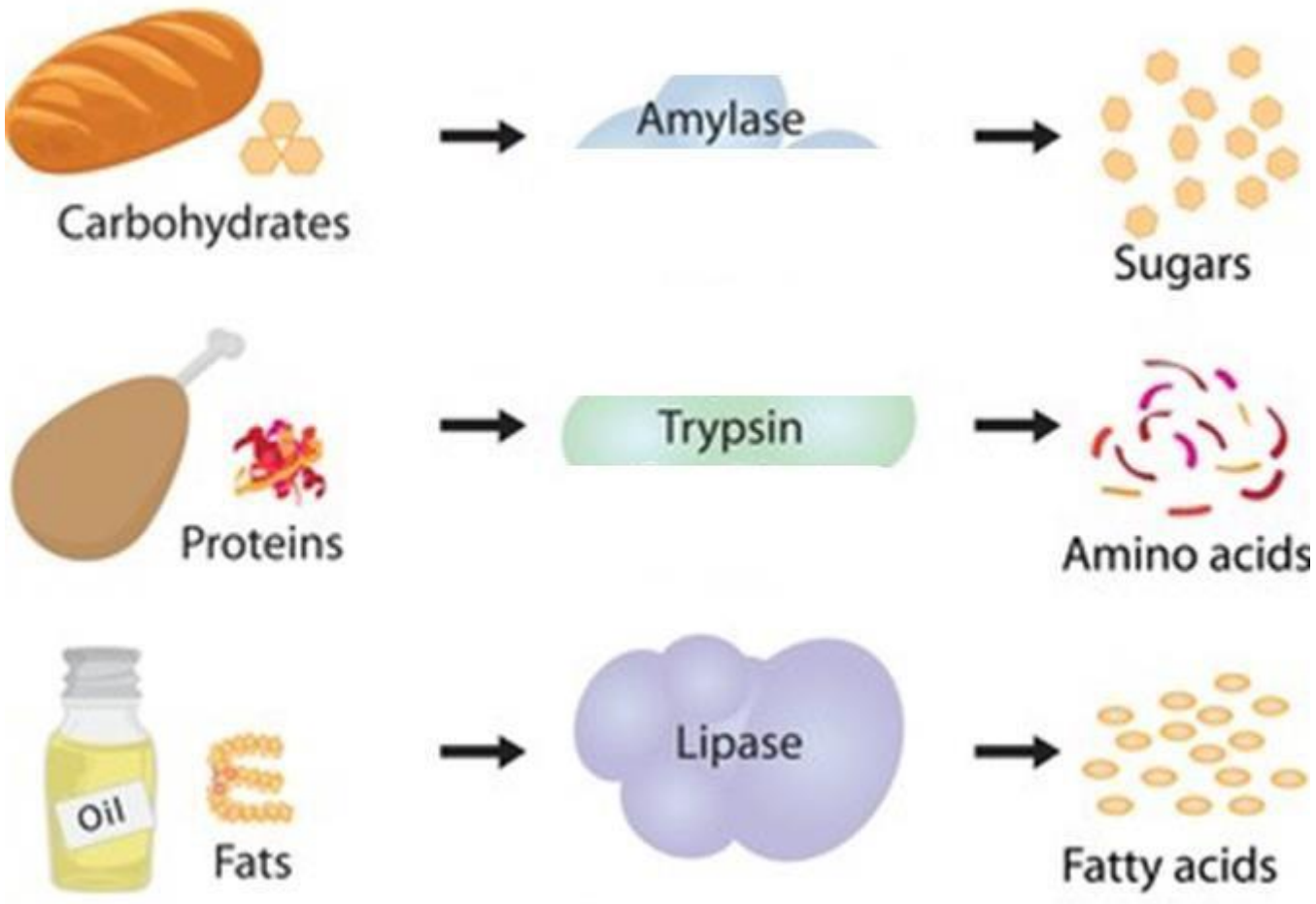
AMINO ACIDS ARE BUILDING BLOCKS OF PROTEINS



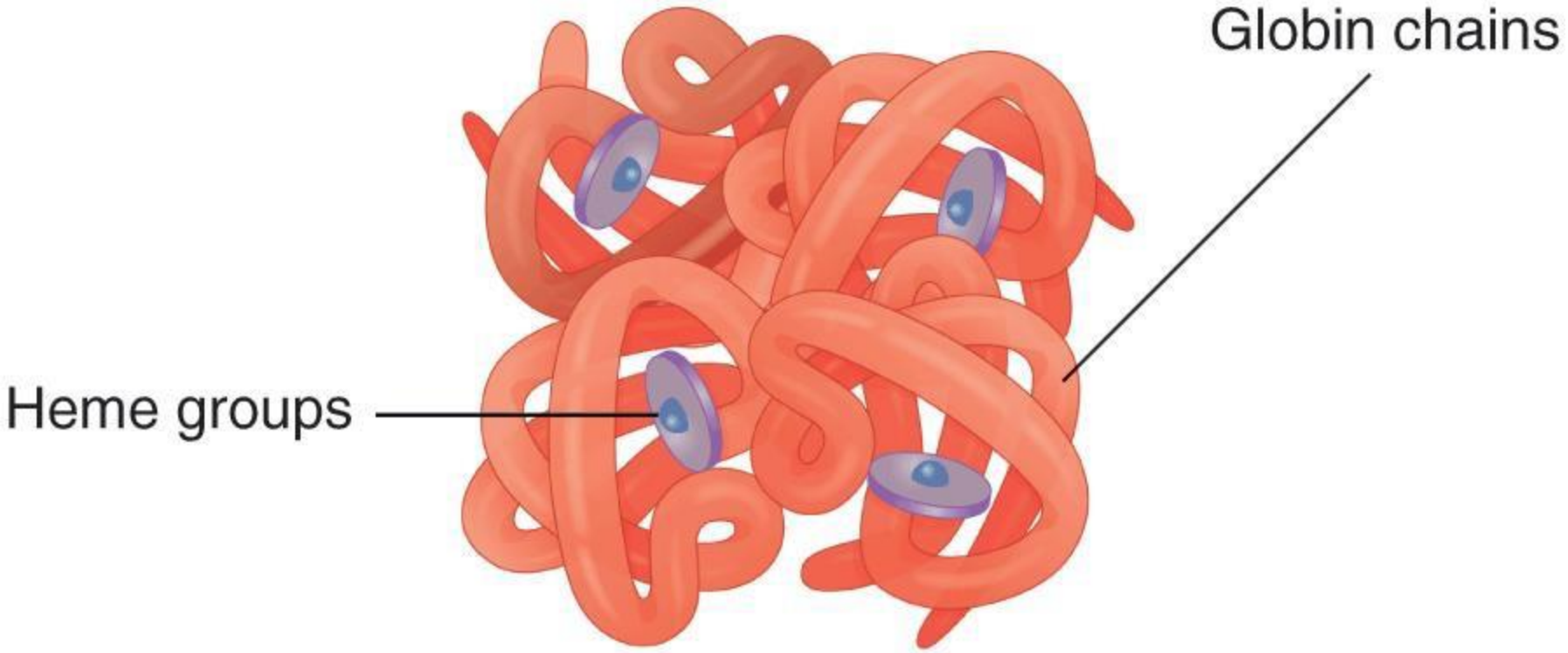
Importance of proteins

- Catalysis (acceleration of chemical reactions in the body) (Enzymes).
- Receptors to enter substances inside the cell (insulin receptor).
- Transport of Oxygen (Hemoglobin = heme + globin).
- Protection against diseases (Antibodies).
- Hormones (Insulin).
- Provide support or strength to your skin, muscles, bones (Collagen).
- Folding of DNA (Histones).
- Some proteins are important for mechanical support (muscles proteins).

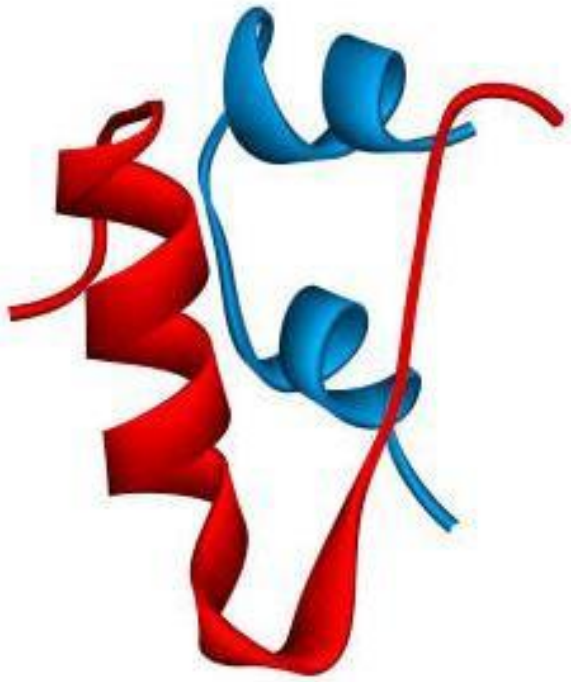
Enzymes are proteins



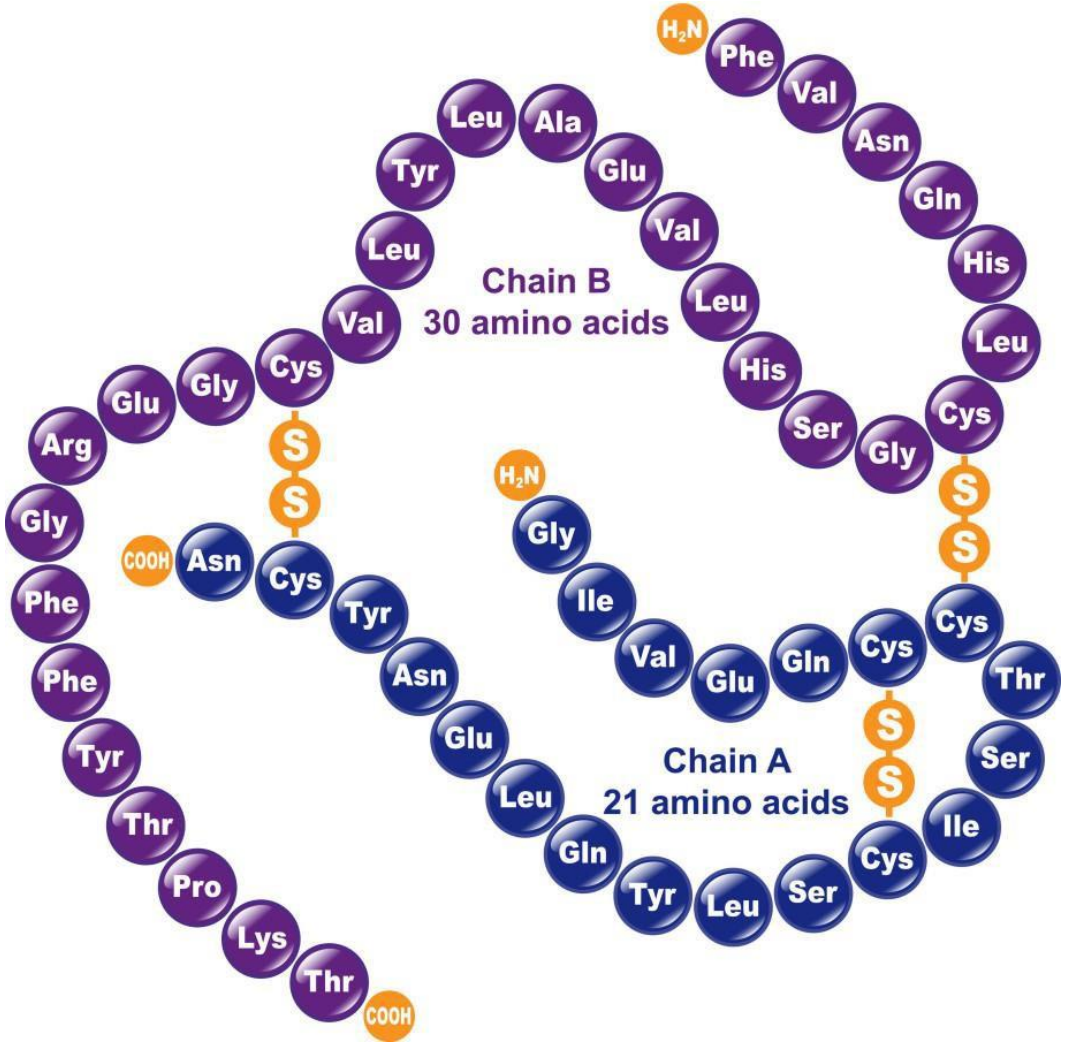
Hemoglobin contains protein



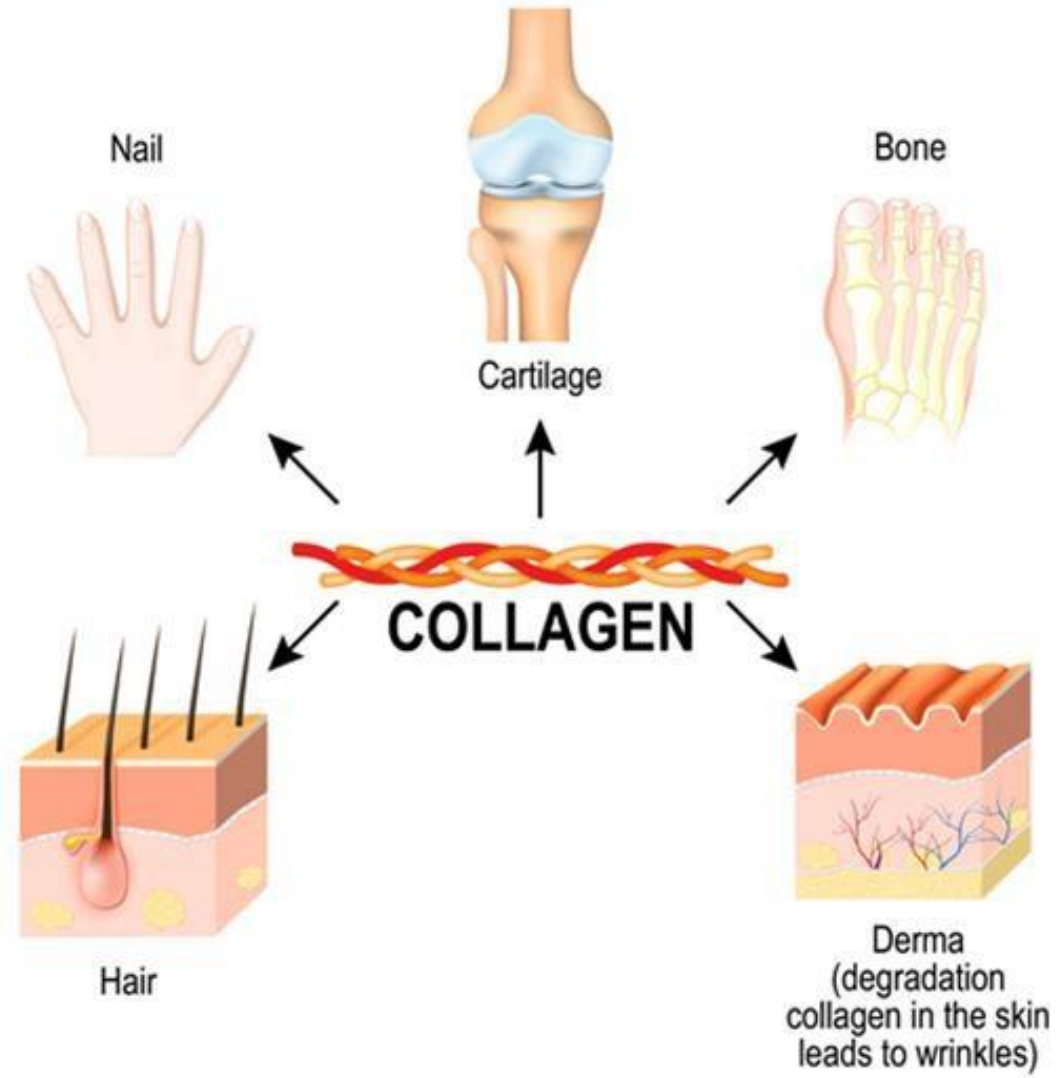
The hormone insulin is a protein



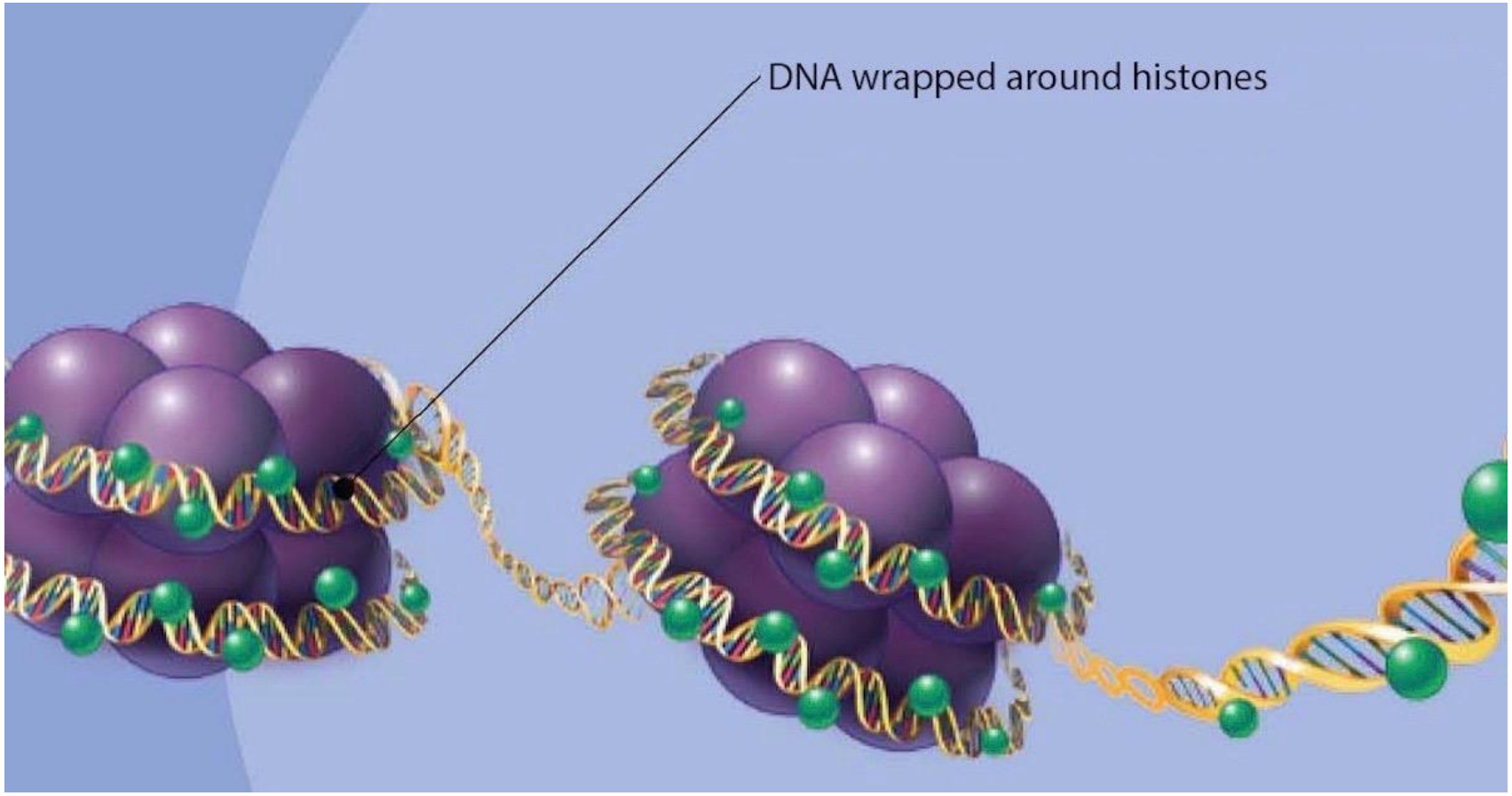
Insulin



Collagen is a protein



Histone is a protein



Amino Acids



- These are the building blocks of proteins (small units that make proteins).
- In our body, there are 20 different types of amino acids.

Amino Acids

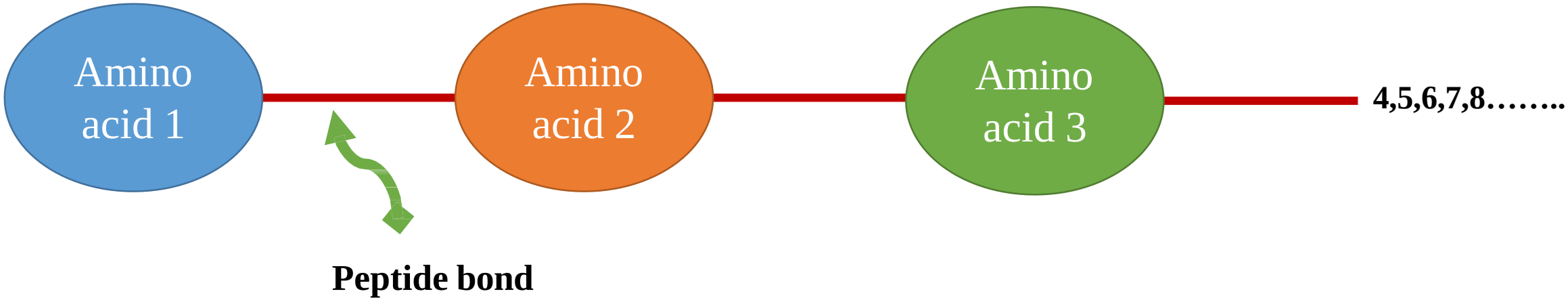


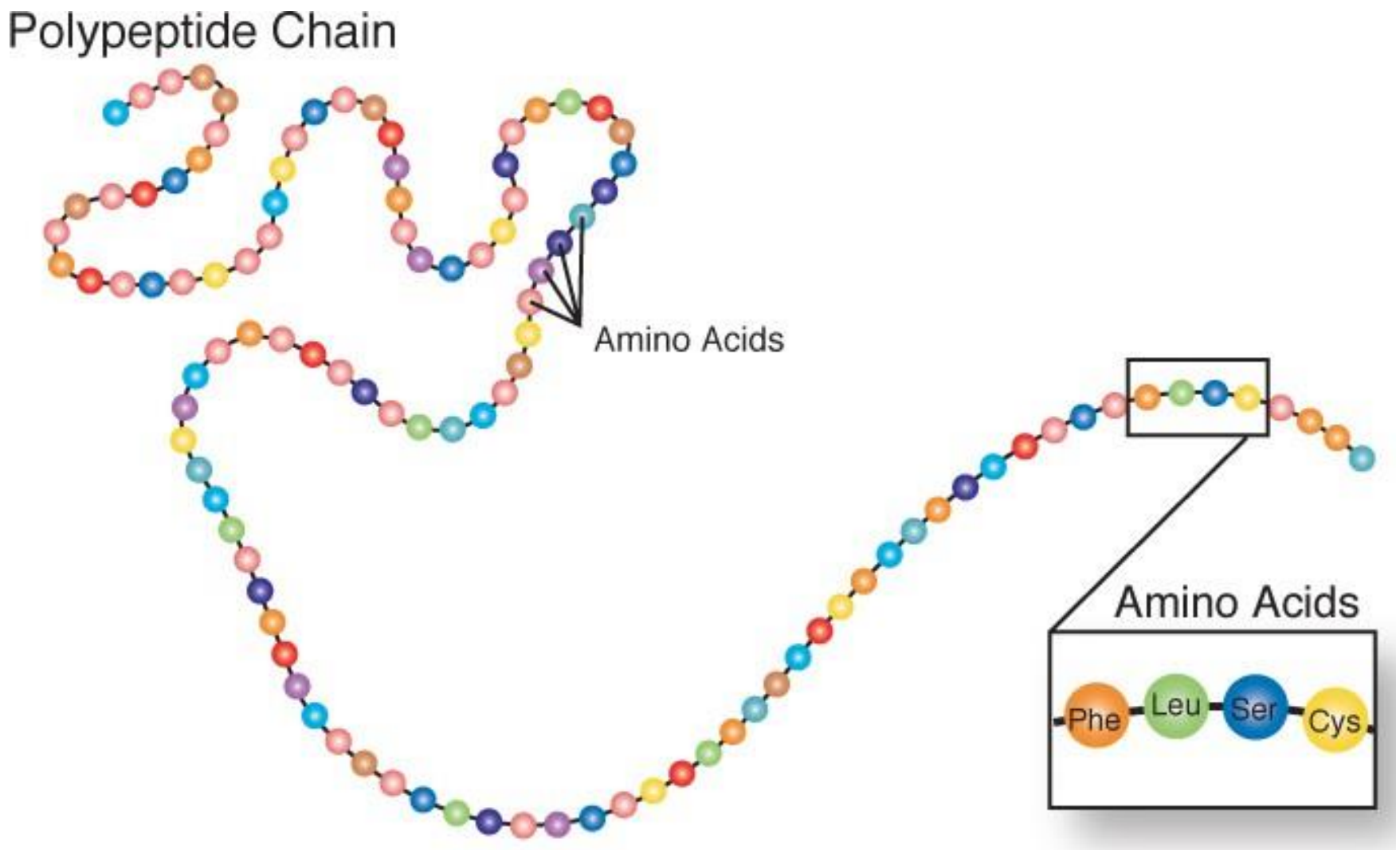
- The bond between amino acids is called peptide bond.
- Proteins differ in the number of amino acids they contain.

Amino Acids

- When two amino acids bind together, they are called dipeptide, When three amino acids bind together, they are called tripeptide.
- More than 10 amino acids together are called polypeptides.

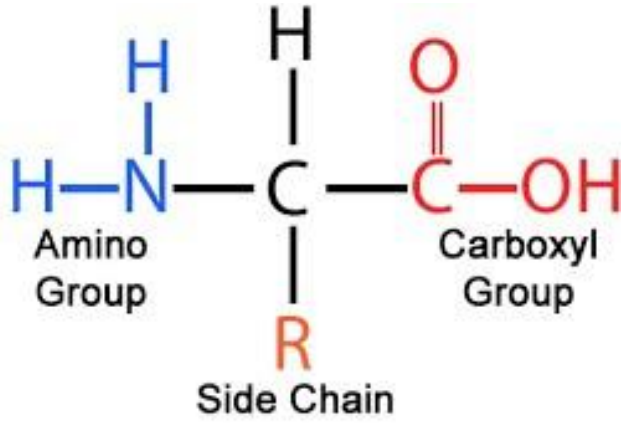




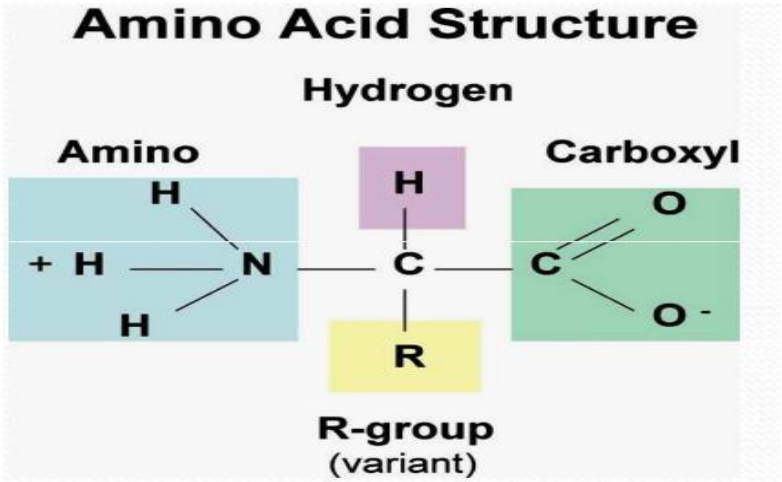

Polypeptide Chain
Amino Acids
Amino Acids
Phe
Leu
Ser
Cys

Structure of amino acids

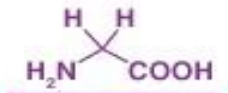
- An amino acid is a molecule that is made up of a carbon atom attached to:
- Hydrogen atom (H)
- Basic amino group ($-NH_2$) or ($-NH_3^+$)
- Acidic carboxyl group ($-COOH$) or carboxylate ($-COO^-$)
- Organic R group (or side chain) that is unique to each amino acid



Amino acid structure

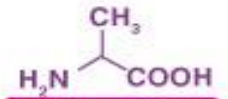


STRUCTURE OF AMINO ACIDS



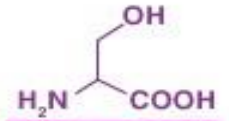
GLYCINE

$C_2H_5NO_2$



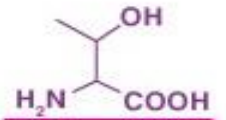
ALANINE

$C_3H_7NO_2$



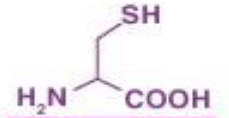
SERINE

$C_3H_7NO_3$



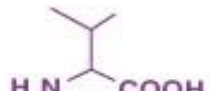
THREONINE

$C_4H_9NO_3$



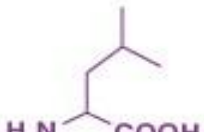
CYSTEINE

$C_3H_7NO_2S$



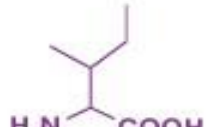
VALINE

$C_5H_{11}NO_2$



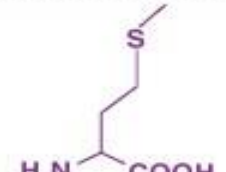
LEUCINE

$C_6H_{13}NO_2$



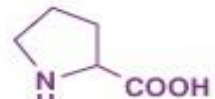
ISOLEUCINE

$C_6H_{13}NO_2$



METHIONINE

$C_5H_{11}NO_2S$



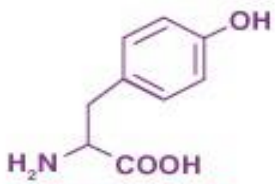
PROLINE

$C_5H_9NO_2$



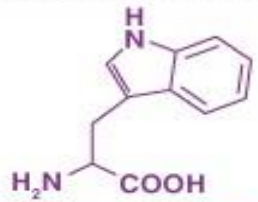
PHENYLALANINE

$C_9H_{11}NO_2$



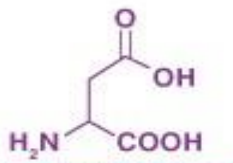
TYROSINE

$C_9H_{11}NO_3$



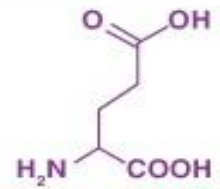
TRYPTOPHAN

$C_{11}H_{12}N_2O_2$



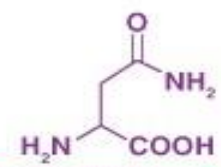
ASPATIC ACID

$C_4H_7NO_4$



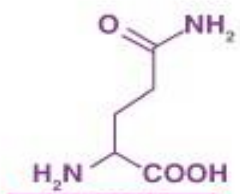
GLUTAMIC ACID

$C_5H_9NO_4$



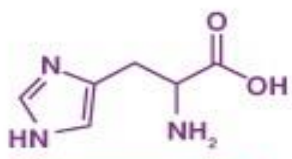
ASPARAGINE

$C_4H_8N_2O_3$



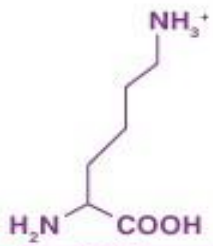
GLUTAMINE

$C_5H_{10}N_2O_3$



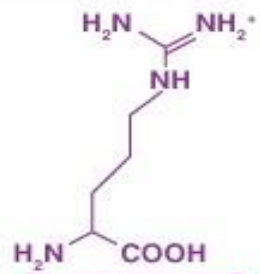
HISTIDINE

$C_6H_9N_3O_2$



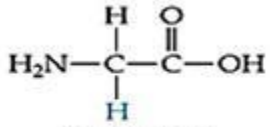
LYSINE

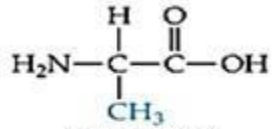
$C_6H_{14}N_2O_2$

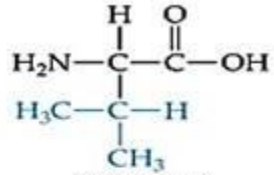


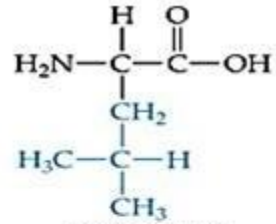
ARGININE

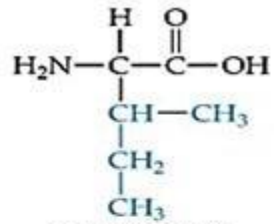
$C_6H_{14}N_4O_2$

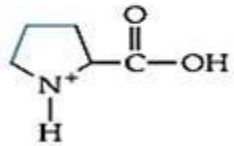

Glycine (Gly)

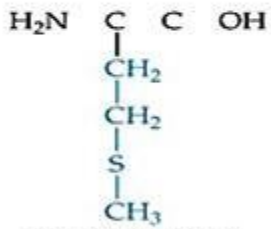

Alanine (Ala)

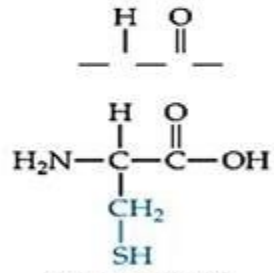

Valine (Val)

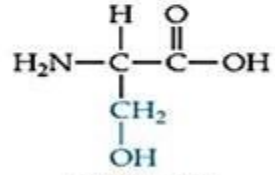

Leucine (Leu)

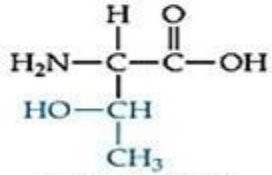

Isoleucine (Ile)

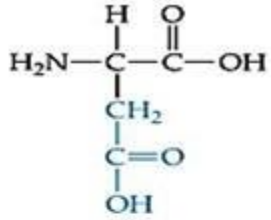

Proline (Pro)

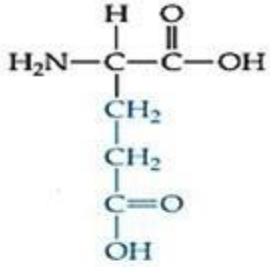

Methionine (Met)

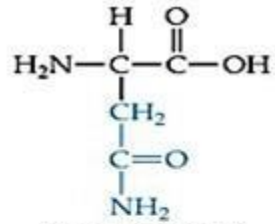

Cysteine (Cys)

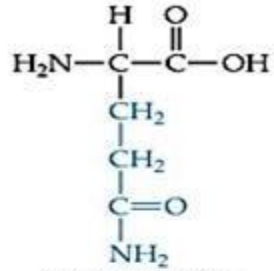

Serine (Ser)

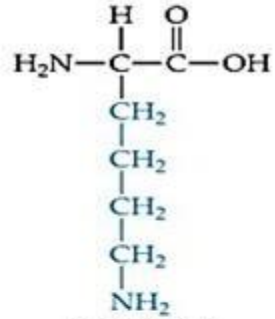

Threonine (Thr)

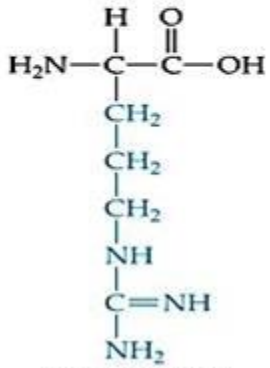

Aspartic acid (Asp)

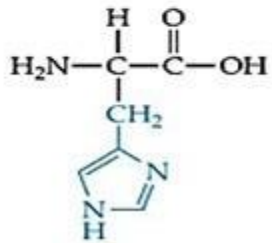

Glutamic acid (Glu)

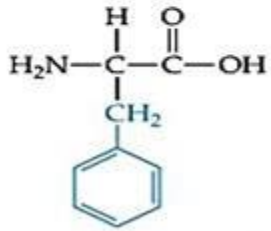

Asparagine (Asn)

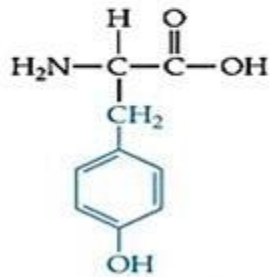

Glutamine (Gln)

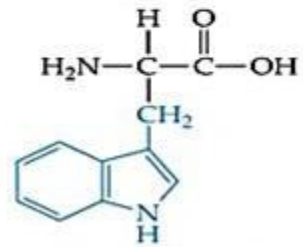

Lysine (Lys)

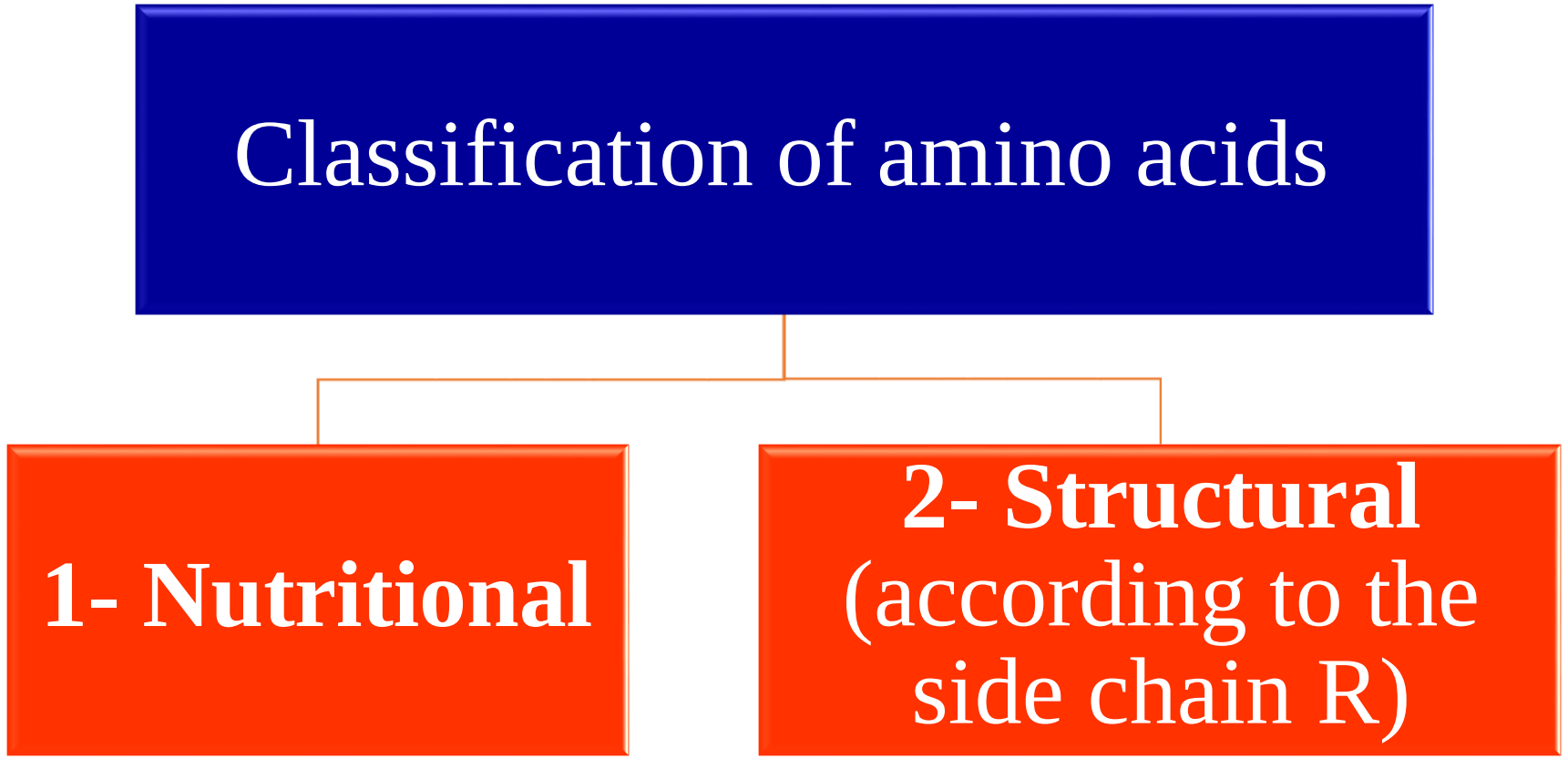

Arginine (Arg)

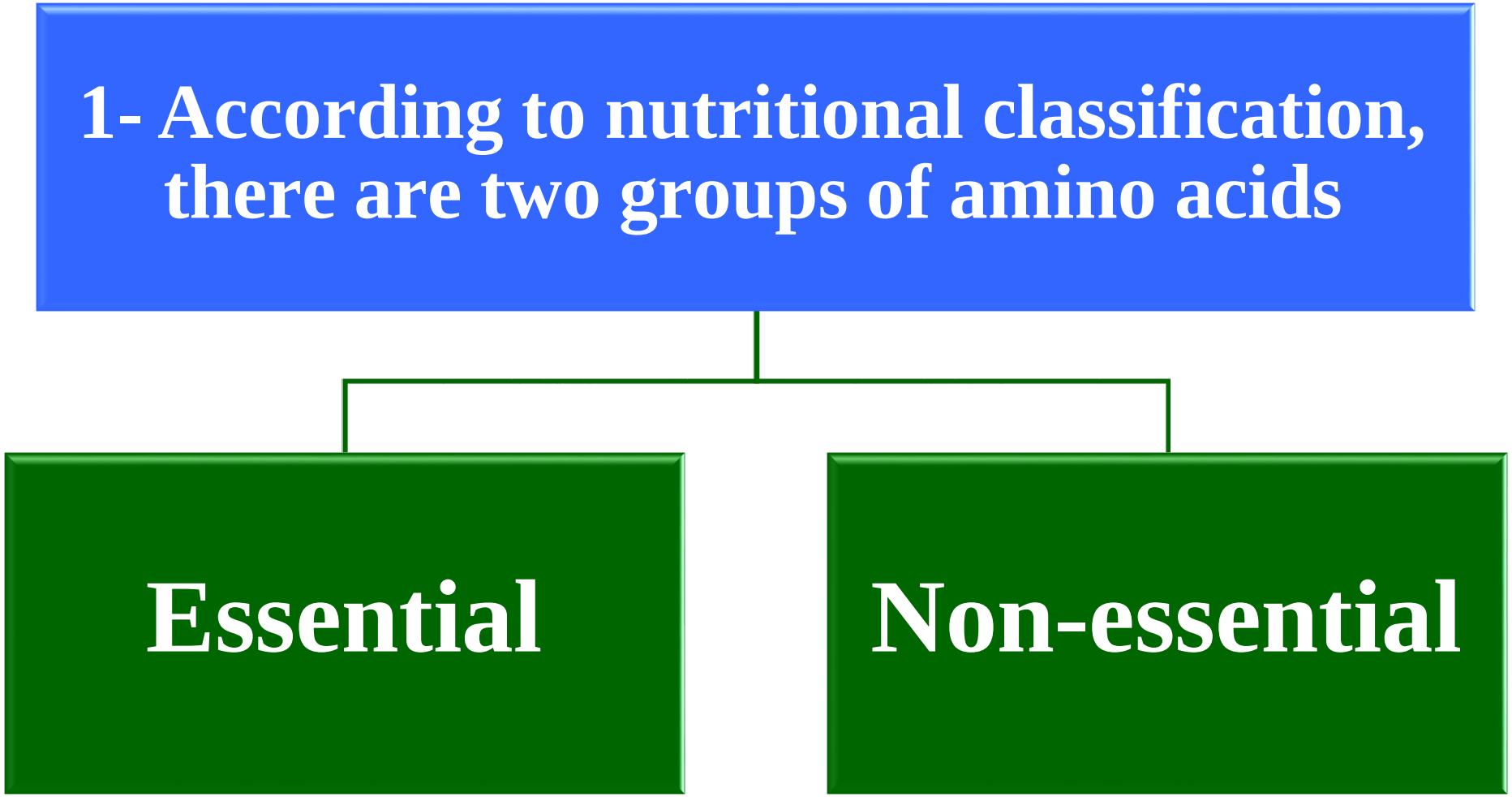

Histidine (His)


Phenylalanine (Phe)


Tyrosine (Tyr)


Tryptophan (Trp)


Classification of amino acids
1- Nutritional
2- Structural (according to the side chain R)


1- According to nutritional classification, there are two groups of amino acids
Essential
Non-essential

Essential amino acids

- They are essential because they cannot be synthesized by human body (nutritionally essential).
- Therefore, these amino acids must be supplied from an exogenous food.
- They are 9 amino acids.

Non-essential amino acids

- Non-essential amino acids are synthesized by the body.
- They are 11 amino acids.

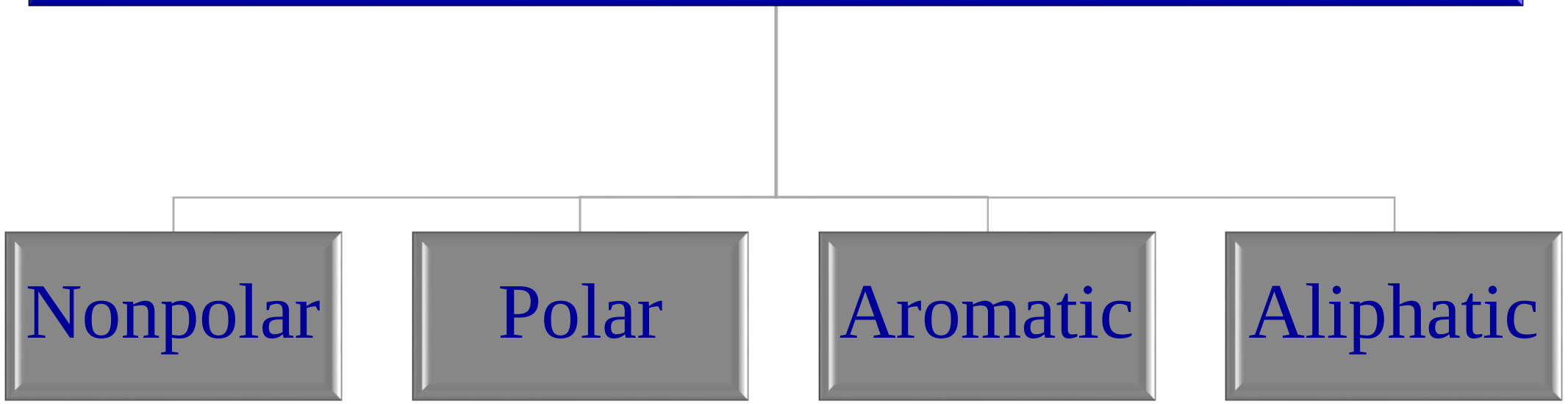
THE 11 NON-ESSENTIAL
AMINO ACIDS

Alanine
Arginine
Asparagine
Aspartic acid
Cysteine
Glutamic acid
Glutamine
Glycine
Proline
Serine
Tyrosine

THE 9 ESSENTIAL
AMINO ACIDS

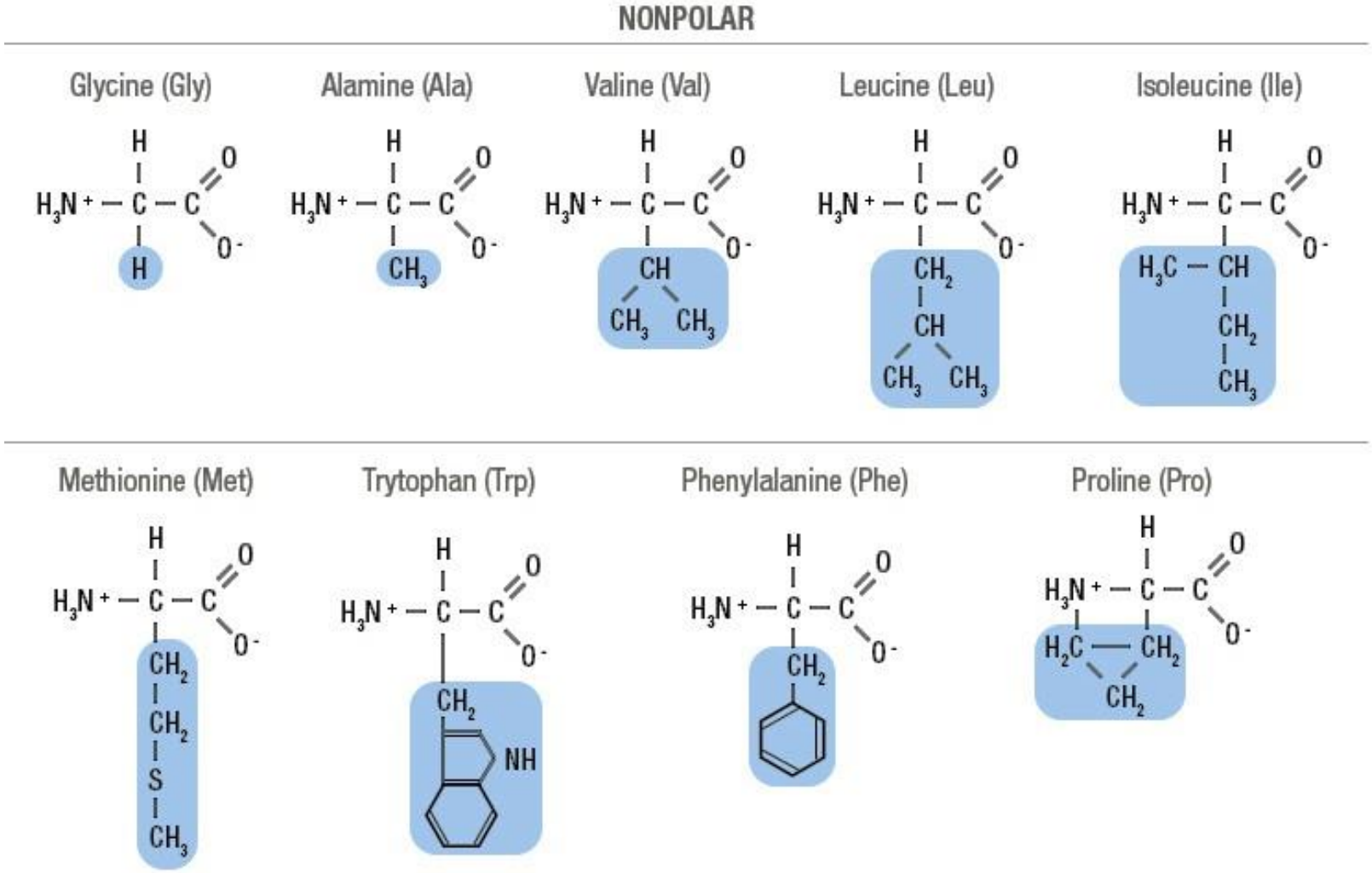
Histidine
Isoleucine
Leucine
Lysine
Methionine
Phenylalanine
Threonine
Tryptophan
Valine

2- According to structure or side chain (R) classification, there are four groups of amino acids



Classification of amino acids according to side chain (structure)

1- Nonpolar amino acids (hydrophobic or less soluble in water):



Classification of amino acids according to side chain

2- Polar amino acids (hydrophilic):

- **They can be:**

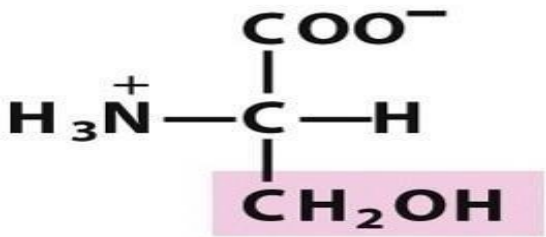
a. **Uncharged (neutral)** contain polar hydrophilic groups (SH,NH_2, OH).

a. **Charged**

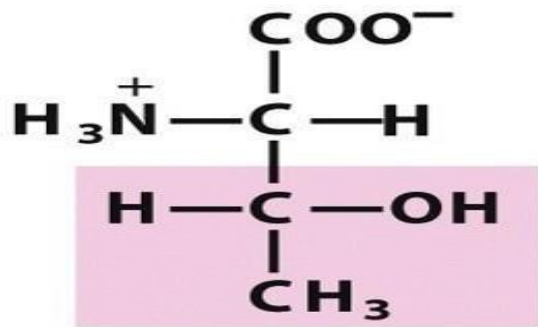
Acidic (negative) **Basic (positive)**

(- charge, have COO^- **in side chain)** **(+ charge, have** NH^+ **group in side chain)**

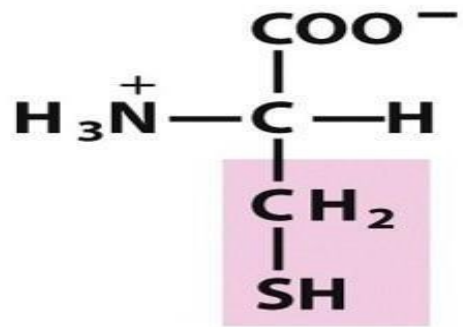
Polar, uncharged R groups



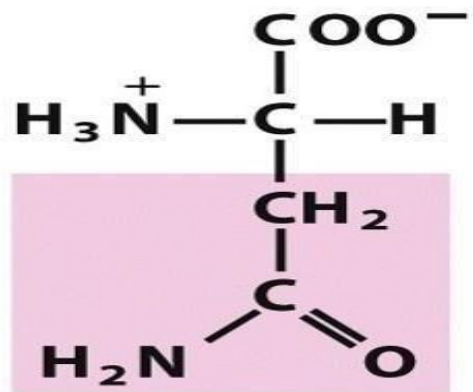
Serine



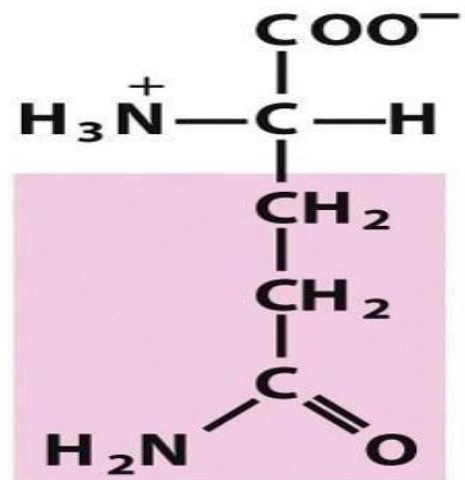
Threonine



Cysteine



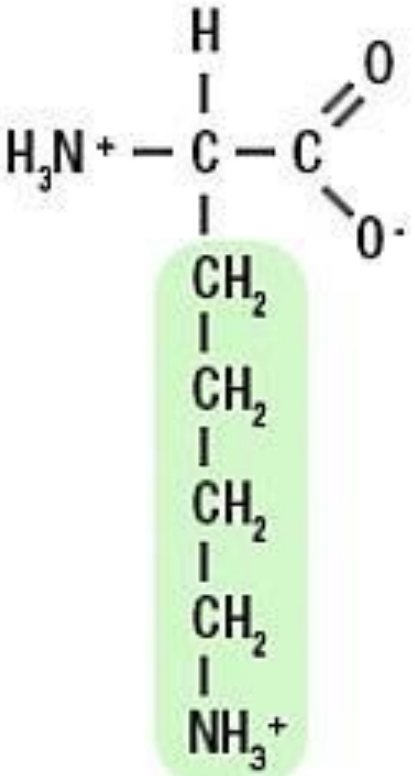
Asparagine



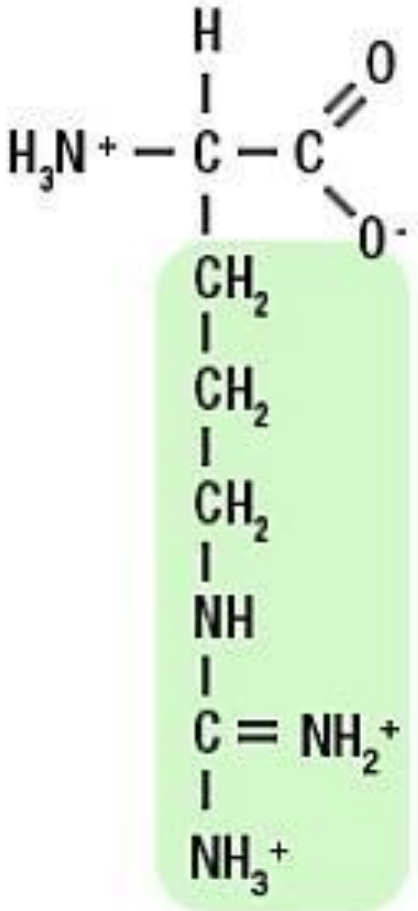
Glutamine

Positively charged R group (Basic amino acids)

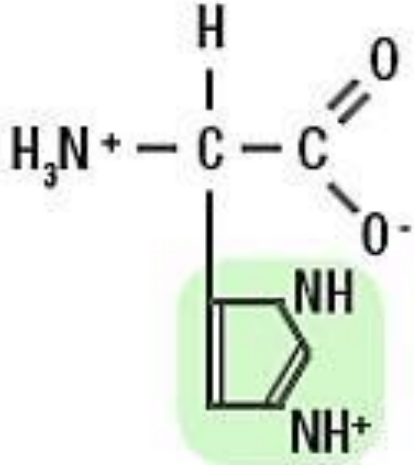
Lysine (Lys)



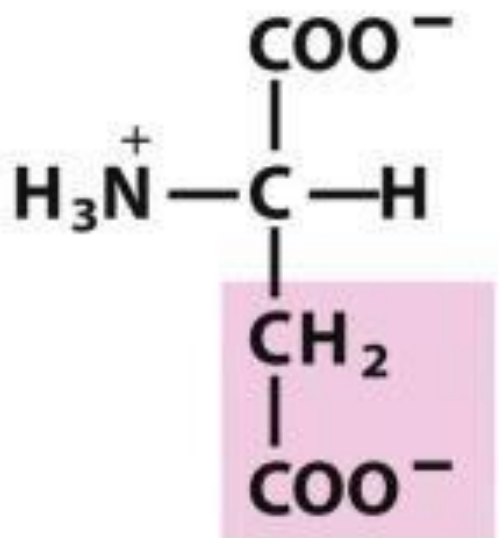
Arginine (Arg)



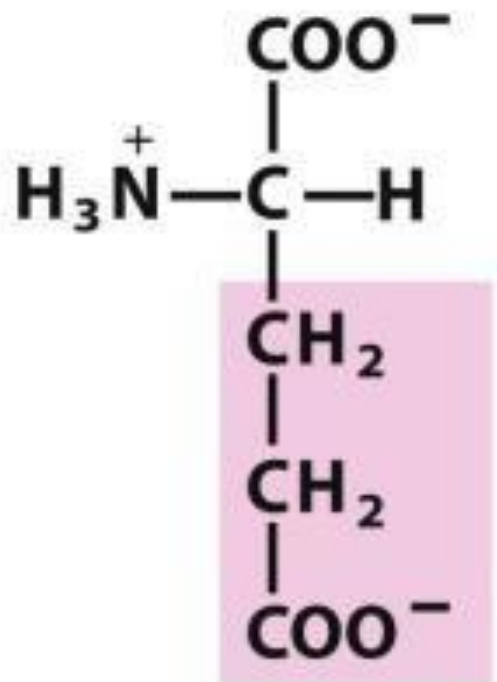
Histidine (His)



Negatively charged R groups



Aspartate

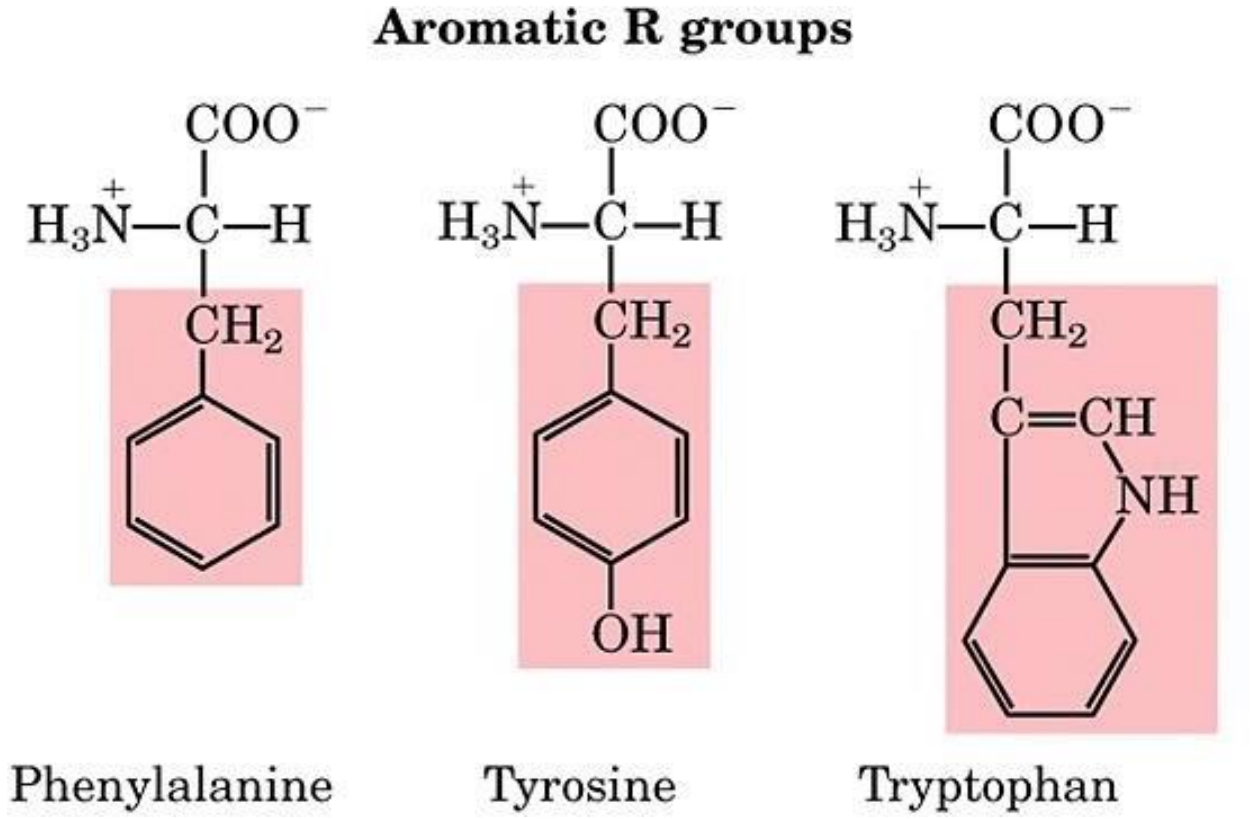


Glutamate

Classification of amino acids according to side chain

3- Aromatic amino acids:

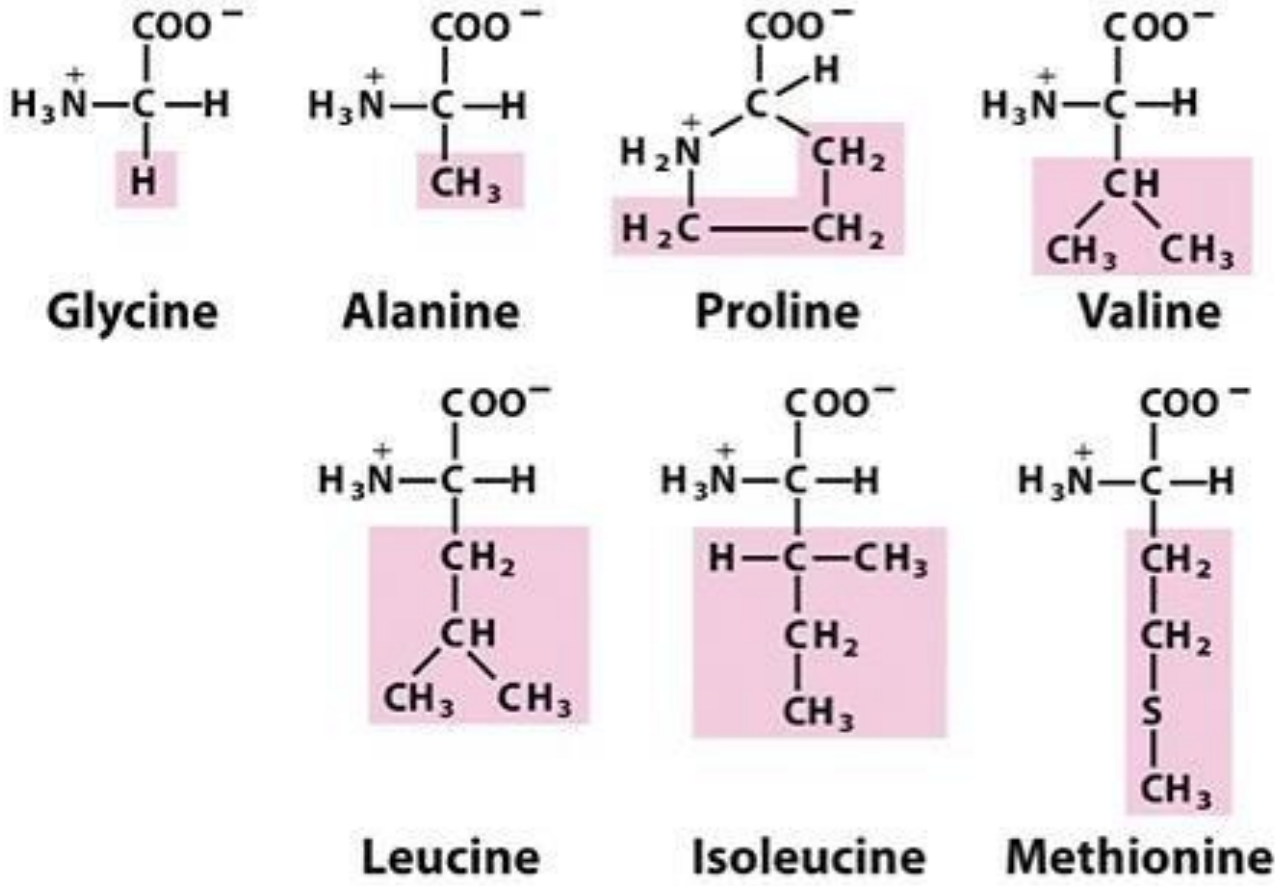
- **Contain aromatic side chain**

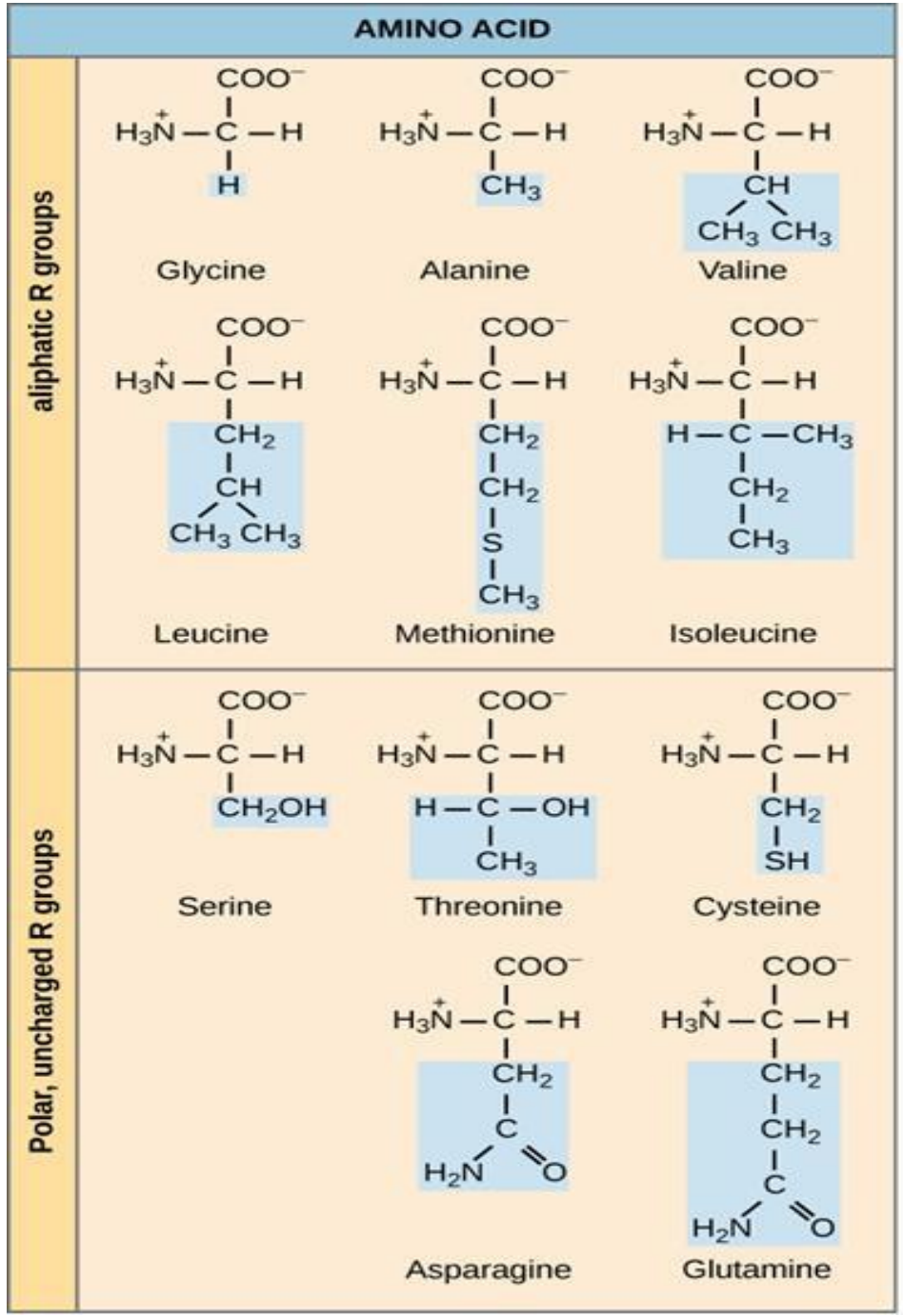


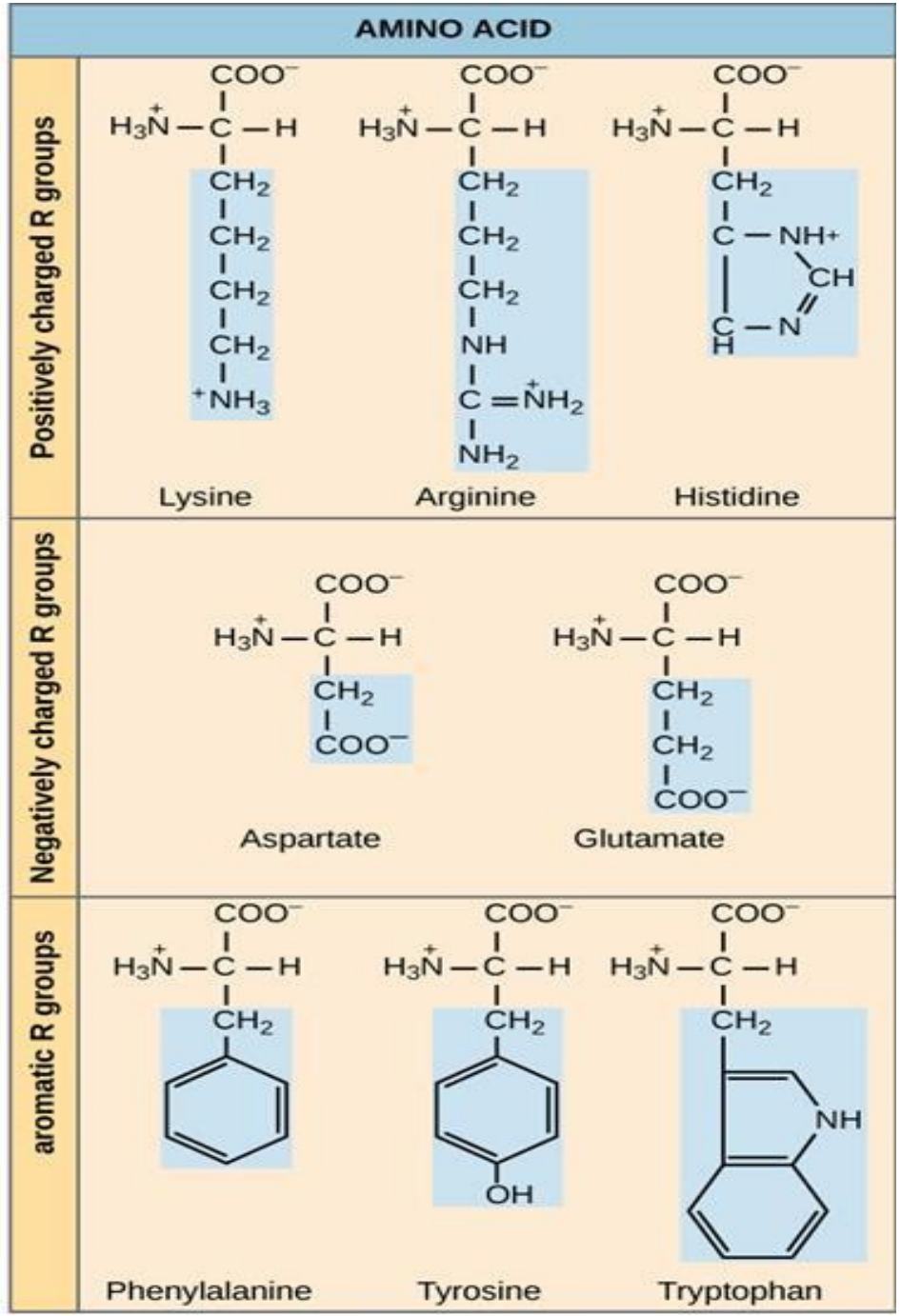
Classification of amino acids according to side chain

4- Aliphatic amino acids:

- **Contain aliphatic side chain**




AMINO ACID
aliphatic R groups
Glycine
Alanine
Valine
Leucine
Methionine
Isoleucine
Polar, uncharged R groups
Serine
Threonine
Cysteine
Asparagine
Glutamine


AMINO ACID
Positively charged R groups
Lysine
Arginine
Histidine
Negatively charged R groups
Aspartate
Glutamate
aromatic R groups
Phenylalanine
Tyrosine
Tryptophan

Questions

➢ State the importance of proteins (four only).

- Catalysis (acceleration of chemical reactions in the body) (Enzymes).
- Receptors to enter substances inside the cell (insulin receptor).
- Transport of Oxygen (Hemoglobin = heme + globin).
- Protection against diseases (Antibodies).
- Hormones (Insulin).
- Provides support or strength to your skin, muscles, bones. (Collagen).
- Folding of DNA (Histones).
- important for mechanical support (muscles proteins).

➢ What is the building block of proteins ?

- Amino acid

Questions

- Number of essential amino acids.
  - 9

- Number of non-essential amino acids.
  - 11

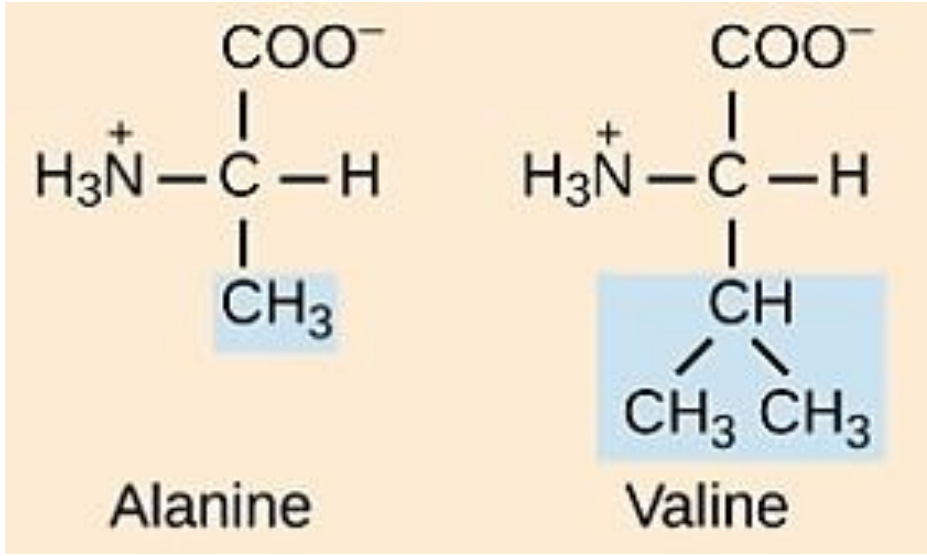
- What is the type of bond between amino acids?
  - Peptide bond

Questions

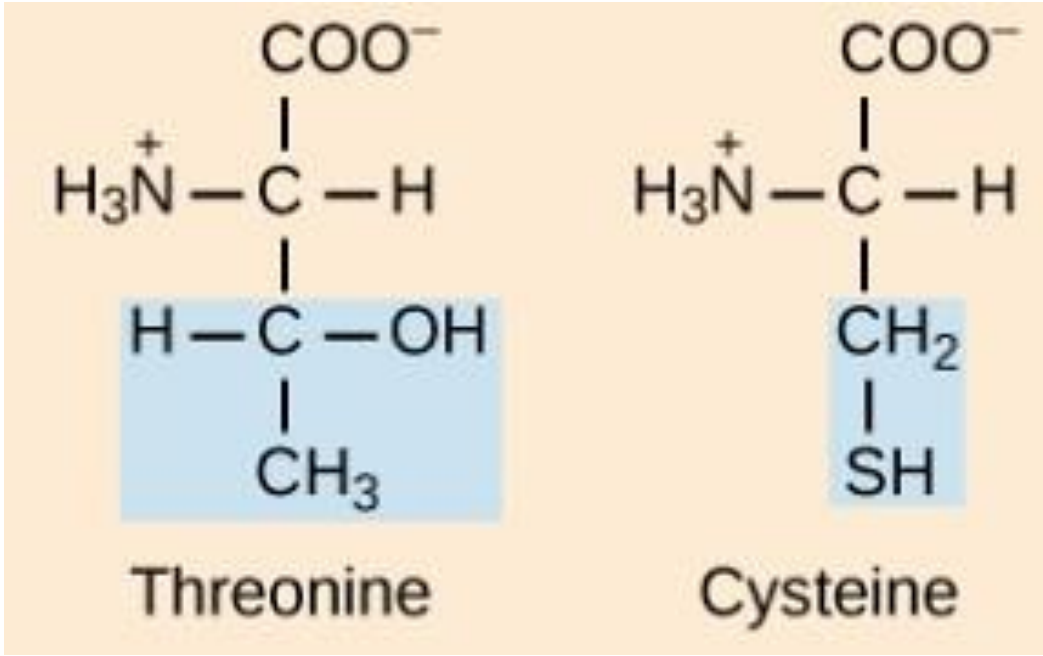
- Mention three of non-essential amino acids.
  - Alanine, arginine, asparagine

- Mention three of essential amino acids.
  - Lysine, leucine, isoleucine

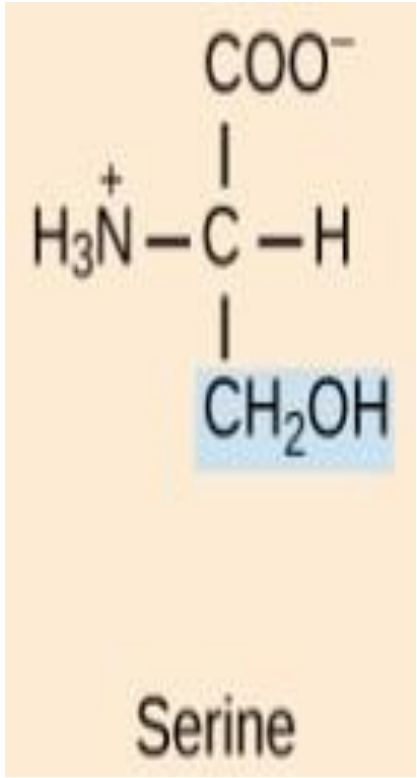
Which of the following is an essential amino acid



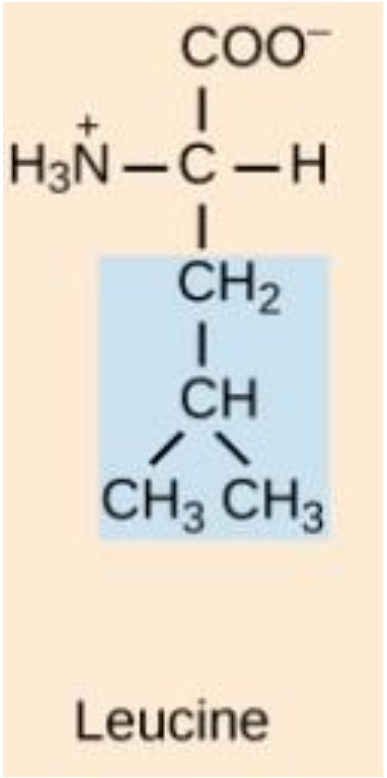
Which of the following is a non-essential amino acid



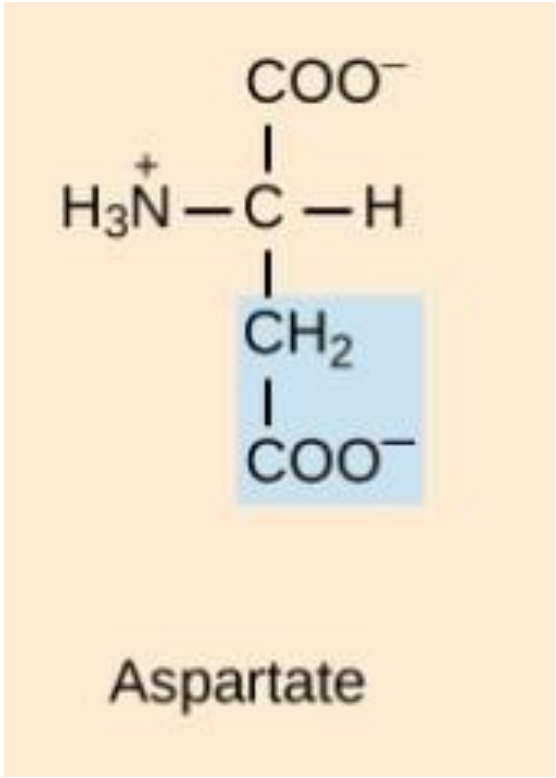
Choose the polar amino acid(s)



Serine

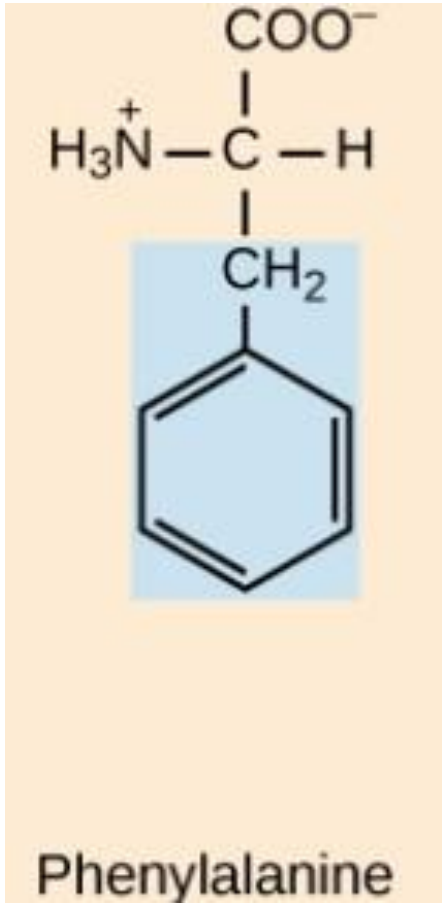


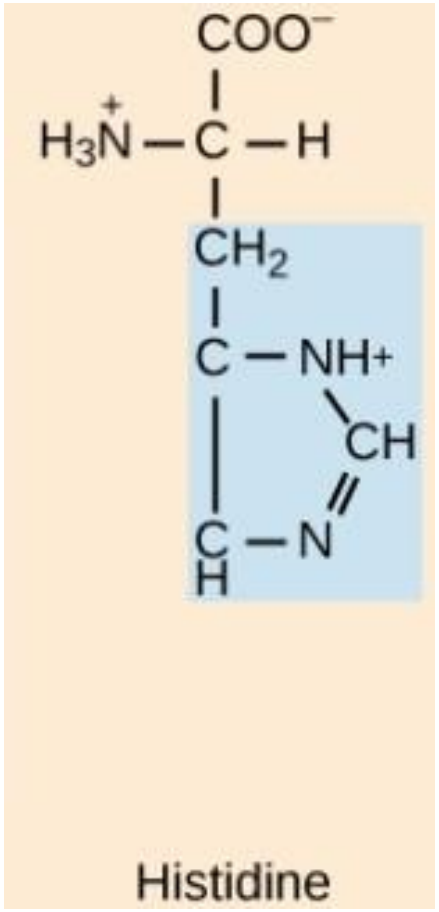
Leucine



Aspartate

Which of the following is an aromatic amino acid ?






GOOD
LUCK!